A 400 kHz, fast-sweep Langmuir probe for measuring plasma fluctuations

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A novel, fast-sweep Langmuir probe has been constructed and successfully operated on ''Thorello.'' It is based on a novel, dual channel circuit that compensates for stray capacitance and permits sweep speeds up to 400 kHz. The circuit response has been tested by measuring the known current–voltage characteristics of resistors and diodes. In addition, the probe has been used to measure the electron temperature and density as well as the plasma potential of plasmas generated in Thorello. A method of three-parameter curve fitting is used to analyze the time-dependent data. The measurements compare favorably with those derived from other standard probe techniques. © *1999 American Institute of Physics*. [S0034-6748(99)03006-3]

I. INTRODUCTION

Electrostatic probes are commonly used to measure density, temperature, and electric potential of low temperature plasmas at the edge of fusion machines, in plasma reactors, in dc-glow discharges and other such devices. The method consists in applying a known voltage to a metallic pin probe immersed in the plasma and measuring the current drawn by it. The great simplicity in determining the current–voltage ($I-V$) characteristics is partially lost by the ambiguous theoretical interpretation of the measurement from which the plasma parameters should be extracted.[1] Nevertheless, the traditional interpolation of the $I-V$ characteristic gives parameters correct within twenty percent. The Langmuir probe measurement has intrinsic good spatial resolution approximated by the electrode size but has questionable temporal resolution.

To gain temporal resolution, researchers make use of triple,[2] multipin,[3] or electronically switched[4] probes for obtaining instantaneous measurements. Some methods, based on curve fitting techniques, use double probes[5] to obtain the correlation between fluctuating parameters, but without temporal evolution. We have adopted the most direct method to sweep the applied voltage (as our presented schema does) fast enough that during a voltage sweep period the plasma parameters remain about constant.[6–8]

In this article we describe a new apparatus for performing fast-sweep probe measurements (Sec. II) which permits us to investigate the electrostatic turbulence commonly observed in the plasma produced by ''Thorello.'' The reliability of the diagnostic apparatus is checked by comparing the results with those obtained with more traditional probe techniques (Sec. III).

This new apparatus gives us the opportunity to study plasma turbulence with clear advantages relative to alternative conventional methods:

(1) to use just one probe tip for doing quasilocal and quasi-instantaneous measurements without having to consider phase delay and shadow problems between different tips,
(2) to associate error bars to the measurements that are obtained by a transparent statistical data analysis,
(3) to accurately measure the probe current in a large range of values,
(4) to evaluate the correlation between temperature, density, and potential fluctuations.[9]

The excellent performances shown by the general purpose components prototype (high current sensitivity, high linearity, large bandwidth, high electron temperature sensitivity, and high reproducibility) make this scheme a viable road for measuring plasma parameters with a temporal resolution good enough to study the turbulent properties of a magnetized plasma.

II. FAST-SWEEP LANGMUIR PROBE TECHNIQUE

A. Basic Langmuir probe theory

Consider a quiescent plasma whose parameters do not change in time and introduce in it a thin conducting wire (a Langmuir probe). Biasing the probe with the voltage V referred to a grounded electrode (like the conducting wall chamber) we can measure the current flowing into the plasma and trace the current as a function of the voltage.

A typical $I-V$ characteristic from Thorello (Fig. 1) presents for negative and positive values of the probe voltage asymmetric saturation currents called ion saturation current I^+ (indicated with the absolute value) and electron saturation current I^-, respectively. The probe current in the ion saturation region has a small negative value (with respect to the electron saturation current) but increases exponentially with increasing voltage, crossing the zero value at the so called floating potential V_f, becoming positive until it reaches the neighborhood of the electron saturation region where there is a characteristic knee at the so called plasma potential V_p.

The values of the local plasma parameters around the probe pin can be estimated by the simplest theory of the Langmuir probe, assuming that the ions are single charge cold particles, the electrons have a Maxwellian velocity dis-

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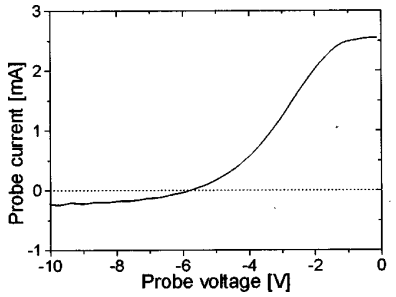


FIG. 1. Typical $I-V$ characteristic of a Langmuir probe immersed in the magnetoplasma produced by Thorello.

tribution, and that a stationary equilibrium is maintained between probe and plasma. The following standard relations appear in Ref. 1:

$$I=I^{+}(e^{e(V-V_f)/kT_e}-1) \quad \text{for } V<V_p, \tag{1}$$

$$I^{+}=\tfrac{1}{2}enA\sqrt{\frac{kT_e}{M}}, \tag{2}$$

$$I^{-}=\tfrac{1}{4}enA\sqrt{\frac{8kT_e}{\pi m}}, \tag{3}$$

and

$$V_f=V_p-\tfrac{1}{2}\left\{\ln\left(\frac{M}{2\pi m}\right)+1\right\}\frac{kT_e}{e}, \tag{4}$$

where e is the electron charge, m the electron mass, M the ion mass, n the plasma density, T_e the electron temperature, V_p the plasma potential, and A is the probe area (in the presence of a strong magnetic field the probe area must be replaced with its geometrical projection transverse to the field lines[1]).

B. Frequency limits

The I–V characteristics can be automatically recorded by sweeping the voltage and measuring the current by an appropriate electronic circuit. Time resolution is obtained if the voltage sweep is continuously performed at a frequency f_{sweep} higher than twice the frequency band f_{fluc} of the fluctuating plasma parameters (Nyquist theorem).

The frequency sweep cannot be too high or otherwise the polarization current[10] in the presence of a magnetic field and the displacement current (both neglected in the above mentioned theoretical model), can alter the probe response. The order of magnitude of the displacement current can be estimated by evaluating the current associated with the sheath probe capacitance C_{sheath} which is given by[7]

$$I_{\text{displ}}\approx C_{\text{sheath}}\frac{dV}{dt}=\epsilon_0\frac{A}{\lambda_D}\frac{dV}{dt}, \tag{5}$$

where λ_D is the Debye length which gives the thickness of the sheath. The sweep frequency where the displacement current effect becomes important is obtained by equating the displacement current and the ion saturation current. The above condition gives the following estimation for this frequency f_{sheath}:

$$f_{\text{sheath}}\approx\frac{\alpha}{2}\frac{kT_e}{eV_0}\omega_{pi}, \tag{6}$$

where α is a constant that is 1 for a ramp voltage sweep and $1/2\pi$ for a sinusoidal voltage sweep. V_0 is the voltage swing for ramp voltage sweep and voltage amplitude for sinusoidal voltage sweep and ω_{pi} is the ion plasma frequency. In the presence of a magnetic field, the perpendicular electric field induced by the probe voltage sweep in the neighboring plasma, leads to an ion polarization drift current given by the following formula:[10]

$$I_{\text{pol}}=A\frac{nM}{B^2}\frac{dE_{\perp}}{dt}\approx A\frac{1}{\Omega_{ci}^2}\frac{ne^2}{M}\frac{d}{dt}\left(\frac{V}{r_{Li}}\right), \tag{7}$$

where Ω_{ci} is the ion cyclotron frequency and r_{Li} is the ion Larmour radius. The ion polarization drift current can contribute to the measured probe current and the frequency f_{pol}, where this current equals the ion saturation current, is given by

$$f_{\text{pol}}=\frac{\alpha}{2}\frac{k\sqrt{T_eT_i}}{eV_0}\Omega_{c,i}. \tag{8}$$

The sweep is usually realized with V_0 about three times T_e, so that f_{sheath} is about a tenth of the ion plasma frequency and f_{pol} about a tenth of the ion cyclotron frequency.

Another problem is to compensate for the stray capacitance due to the electrostatic coupling between the probe pin and the BNC cable. The coupling with the conductive wall chamber is negligible. The stray capacitance adds to the input electronic circuit A contribution to the measured current given by

$$I_{\text{stray}}=C_{\text{stray}}\frac{dV}{dt} \tag{9}$$

which becomes comparable to the ion saturation current at the following frequency sweep:

$$f_{\text{stray}}=\frac{\alpha}{2}\frac{nA}{CV_0}\sqrt{\frac{kT_e}{M}}. \tag{10}$$

This last problem can be solved by using a second probe (reference or dummy probe) placed outside the chamber,[11] so that it is not exposed to the plasma. The same voltage sweep must be applied to the active and dummy probes and the currents measured from both probes. The difference between the two currents is the true measure of the current flowing in the plasma. This differential measurement is capable of reducing the equivalent stray capacitance and any common mode noise that influence the two probes.

Finally for reliable fast-sweep Langmuir probe measurements the operation frequency of the apparatus must satisfy the relations:

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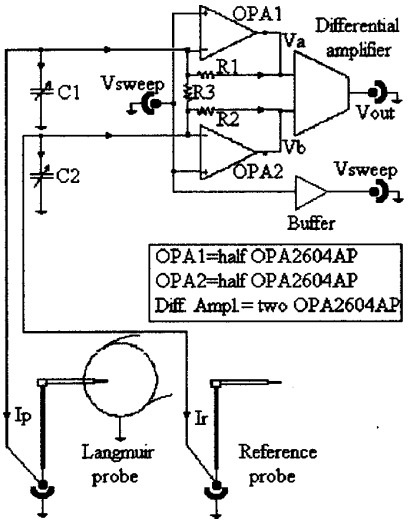


FIG. 2. Block diagram of the fast-sweep circuit connected to a Langmuir probe immersed in a plasma and to a reference or dummy probe outside the plasma.

$$2f_{\max} \leqslant f_{\text{sweep}} \ll f_{\text{stray}}, f_{\text{sheath}}, f_{\text{pol}}. \tag{11}$$

C. Experimental setup

It is relatively easy to sweep the probe voltage at a high frequency but quite problematic to measure correctly the current absorbed by the load. Traditional schemes introduce a shunt resistor in series and then measure the voltage drop. The shunt resistor introduces an error in the voltage sweep, in other words it gives a swing on the probe voltage according to the plasma current response. Moreover, the swept voltage applied to the dummy probe cannot be the same as that applied to the active probe. This is because the voltage drops over the shunt resistors depend on the current flowing and this difference cannot be compensated even though the stray capacitance of the two probes are exactly matched.

In Fig. 2, the block diagram of the electronic circuit that solve these problems is shown. The Langmuir probe and the dummy probe are virtually and independently connected to the same signal generator by two operational amplifiers. This solution guarantees that the voltage applied to the probes are the same and independent on the currents drawn, and at the same time the two probes do not interfere with each other. The current drawn by the probe must pass only the corresponding feedback resistor, in fact the operational amplifier input impedance is almost infinite [the chosen operational amplifier OPA2604AP are high frequency, high voltage, input field-effect-transistor (FET) operational amplifiers]. The voltage signals V_a and V_b to the output of the two input stages are given by

$$V_a = V_{\text{sweep}} + \left(C_1 \frac{dV_{\text{sweep}}}{dt} + I_p \right) R_1, \tag{12}$$

$$V_b = V_{\text{sweep}} + C_2 \frac{dV_{\text{sweep}}}{dt} R_2, \tag{13}$$

where I_p is the plasma current, C_1 and C_2 are the total input capacitances (given by the sum of the probe capacitance, the cable capacitance and the on board adjustable capacitance) and R_1 and R_2 are the feedback resistors. A high precision differential voltage amplifier stage with a gain G follows so that the output voltage is

$$V_{\text{out}} = G\left[(C_1 R_1 - C_2 R_2) \frac{dV_{\text{sweep}}}{dt} + I_p R_1 \right], \tag{14}$$

if the following condition is matched:

$$C_1 R_1 = C_2 R_2. \tag{15}$$

V_{out} is proportional to the plasma current I_p with a gain factor R given by the product of the feedback resistance R_1 and the differential amplifier gain G:

$$V_{\text{out}} = R_1 G I_p = R I_p. \tag{16}$$

The relatively small resistance R_3 connecting the two inputs, does apparently not influence the circuit because the two nodes are at the same voltage, being virtually connected to V_{sweep}. Instead, R_3 is important for balancing any mismatch between the two input channels due to the manufacturing processes, improving the sensibility, reducing the input equivalent noise, and increasing the global response time.

The first important aspect of this circuit is that it exhibits a natural quasi-zero impedance on its input nodes like an almost ideal differential current amplifier. This property is to the knowledge of the authors not shared with the schemes presented in the literature (Refs. 6–8). The second important aspect is that the circuit power supply, the probe sweep voltage, and the measured current are all referred to the wall chamber ground giving rise to a small ground loop noise. The critical point of the circuit is the stability that can be lost if the equivalent input capacitance of the circuit is too high. We successfully realized a stable circuit connecting the probes and the operational amplifiers input by a BNC cable about one meter long (about 100 pF cable capacitance).

D. Features

A preliminary fine tuning of the apparatus without a plasma discharge must be done before the actual measurements in the plasma. This is needed in order to match the operational amplifier feedback resistances and the probes stray capacitance. The tuning is done by applying the voltage sweep to the probes and adjusting the input variable capacitance C_1 and C_2 together with the feedback resistance in order to reach a condition where the measured current is zero. Another way of tuning the system is to put a known resistance load R_{sample} at one input and then adjust like before C_1, C_2, and the feedback resistance to reach a condition where the output voltage satisfies Eq. (16) with $I_p = V_{\text{sweep}}/R_{\text{sample}}$.

This procedure can also be used for testing the characteristic electrical features of the apparatus, and the results of this test are shown in Table I.

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TABLE I. Principal electrical properties of the fast-sweep Langmuir probe prototype shown in Fig. 2 and built in our laboratory.

Features	Value
Half gain bandwidth	800 kHz
Adjustable gain	0.1–10 kΩ
Non linearity	<1%
T_e resolution	≤10 meV
Equivalent input current noise	≤4 μA

In the presence of plasma the sweep voltage and the output current are simultaneously sampled by means of a digital oscilloscope and stored on a magneto-optical disk for subsequent data reduction analysis.

Figure 3 shows the time series of the two signals as sampled in the magnetoplasma produced by Thorello[12] with the Langmuir probe positioned 8 cm from the wall chamber. Thorello is a toroidal device with minor radius $a=8$ cm, major radius $R=40$ cm, and maximum toroidal magnetic field $B=2$ kGauss. The plasma is produced by a steady state discharge between one or more incandescent tungsten wires placed in the inner part of the torus and the chamber walls with a polarization voltage of about $V_f=-100$ V. The typical plasma parameters are the electron temperature $T_e=1$ eV, the density $n=10^{11}$ cm^{-3}, and the neutral gas pressure $P_n=10^{-4}$ mbar.

In Fig. 4 details of a single sweep are shown: it is possible to see the derivative behavior of the circuit response to a large discontinuous signal (the ramp voltage drop) which shows high frequency (3 MHz) oscillations (dumped in 2 μs) due to the critical stability of the circuit imposed by the requested large bandwidth. This behavior can be improved by a more accurate electronic design based on more expensive commercial electronic chips or decreasing the BNC cable length.

E. Data reduction analysis

The single $I-V$ characteristics can be fitted in the chosen voltage window by the Langmuir curve to extract the plasma parameters as predicted by theory (Fig. 4). An automatic fitting procedure has been developed (based on the Fortran CERN package subroutine called MINUIT), which minimizes the chi-square function:

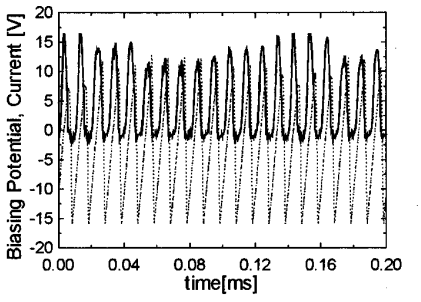


FIG. 3. Typical sweep voltage (...) and output signal (−) obtained by the fast-sweep Langmuir probe with a sweep frequency of 100 kHz. Here, the current is multiplied by an R factor gain of 5 kΩ.

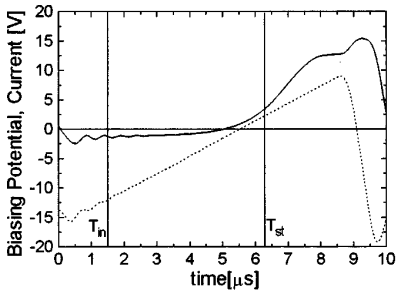


FIG. 4. Details of a typical fast-sweep Langmuir probe $I-V$ characteristic of Fig. 3 (the current is multiplied by an R factor gain of 5 kΩ). The fit is performed in the selected time window $[T_{in},T_{st}]$.

$$\chi^2=\sum_{j=N_{in},N_{st}}\frac{(I_j-I(a_1,a_2,a_3;V_j))^2}{\sigma^2}, \tag{17}$$

where the fit starts with the N_{in} data sample and stops at N_{st}, I_j is the jth sampled current value, $I(a_1,a_2,a_3,V_j)$ is the fitted curve evaluated at the jth sampled voltage value V_j, a_k $(k=1,2,3)$ are the fitting parameters, and σ is the error of the measured current. The error σ can be assumed to be the root-mean-square between the fast-sweep current noise (Table I) and the 8-bit digitalization error introduced by the oscilloscope. Good fits are selected accepting only those which have a chi-square value to degree of freedom $\chi^2_{\mathrm{DOF}}=\chi^2/\mathrm{DOF}$ less than 0.7, where the data degrees of freedom are defined by the formula $\mathrm{DOF}=N_{st}-N_{in}-3$, (each fitting parameter reduces the DOF by one). For example if DOF $=100$, the confidence level is about 99%.[13] The fitting Langmuir curve is linear in two parameters if we chose $a_1=I_{si}$ and $a_2=I_{si}\exp^{-eVf/kTe}$ as fitting parameters, but is inevitable not linear in the third one, $a_3=kT_e/e$. The error Δa_k on the kth parameter is given by the variation that increases the chi-square of one from the minimum value. This corresponds to a bounded region in the three-dimensional parameters space centered around the fitted parameters where the fit remains statistically significant.

F. Preliminary test

The functionality of the system has been verified by inserting a Germanium diode parallel to the active probe when the discharge was turned off and the $I-V$ characteristic of

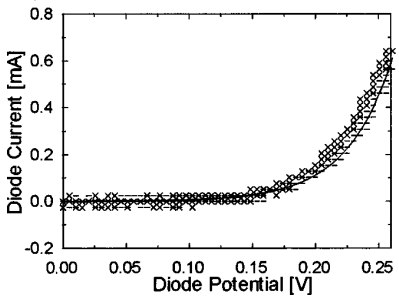


FIG. 5. The current–voltage characteristic of a Germanium diode measured employing a digital ammeter and a digital voltmeter (x), by the fast-sweep probe (−) and the fitted curve ($T_e=0.035\pm0.010$ eV and $\chi^2_{\mathrm{DOF}}=0.05$) obtained from the Langmuir function (full line).

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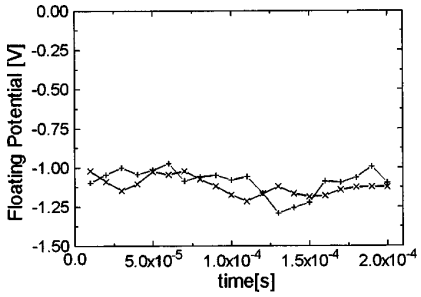


FIG. 6. The temporal evolution of the floating potential in the magnetoplasma produced by Thorello at 6 cm from the conductive wall for $B = 0.15$ T as measured simultaneously by the fast-sweep probe (+) and by a nearby floating probe (x).

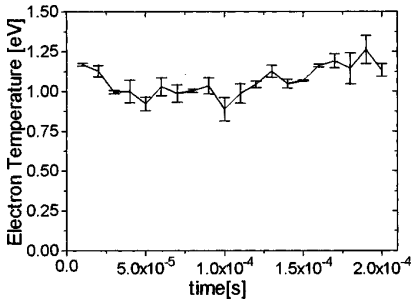


FIG. 8. The temporal evolution of the electron temperature in the magnetoplasma produced by Thorello at 6 cm from the conductive wall for $B = 0.15$ T as measured by the fast-sweep probe.

the diode reconstructed by the fast-sweep apparatus running at 400 kHz. The stored curve has been compared with a direct dc measurement and the two plots look very similar as can be observed in Fig. 5. The characteristic of a diode is described by a Langmuir curve with V_f as the diode voltage threshold, I_{si} the diode inverse current, and T_e the environment temperature (about 0.025 eV) multiplied by a constant of order one dependent on the diode. The best fit from the fitting procedure gives for the parameter T_e a value of about 0.035 ± 0.010 eV. The fit results for I_{si} and V_f are not significant because the inverse current is immersed in the electronic noise, in fact the inverse current value for a Ge diode is expected to be about 1 μA. This means that the apparatus at 400 kHz gives a correct response to a load having a Langmuir like $I-V$ characteristic and moreover the fast-sweep electron temperature sensitivity is less than about 10 meV (Table I). The repeatability of the electron temperature measurement is estimated by measuring 1000 $I-V$ characteristics of different type of diodes (Germanium, Silicon and LED diode) and plotting the dispersions of the fitted T_e electron temperature parameter. The result is that the system presents intrinsic fluctuations of about 1.5% in the T_e parameter of the fitted curves.

III. MEASUREMENTS AND COMPARISONS

The measurements have been obtained by a moving radial probe made with three parallel and cylindrical tungsten wires. The wires have a diameter of 0.3 mm and are shielded by alumina pipes with 7-mm-long tips exposed to the plasma. Keeping tip 1 as the geometrical reference point, tip 2 is poloidaly separated from tip 1 by a distance $d_{2,1} = 6$ mm, and tip 3 is toroidaly separated from tip 1 by the distance $d_{3,1} = 5.5$ mm. The fast-sweep probe apparatus has been connected to tip 1 and used to measure the floating potential, the ion saturation current and the electron temperature for different values of the distances from the minor axis of the chamber. Each data set consists of 65 $I-V$ characteristics, reconstructed with 500 sampled points and recorded by a 100 kHz ramp sweep. The voltage amplitude is set to about $3T_e$ in such a way as to reach the ion saturation current region but not the electronic saturation region in order to avoid perturbing the surrounding plasma or influencing the response of the other tips. The sweep frequency is high enough to follow the fluctuations which typically have high frequency components below 50 kHz. In fact we experimentally observed by the triple probe method (see below) that in the chosen 10 μs sweep time window the plasma parameters change less than about 1.5%, so that the plasma can be supposed ''frozen'' in time on this time scale. The measured parameters as extracted from the $I-V$ characteristics by the minimum least square method are then compared with the results obtained from the simpler traditional methods.

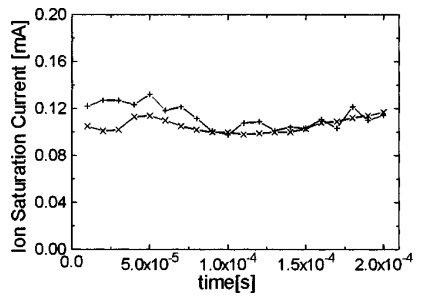


FIG. 7. The temporal evolution of the ion saturation current in the magnetoplasma produced by Thorello at 6 cm from the conductive wall for $B = 0.15$ T as measured simultaneously by the fast-sweep probe (+) and by a nearby probe polarized in the ion saturation mode (x).

A. Floating potential and ion saturation current measurements

To make sure that the fast-sweep probe retains the right development in time of the fluctuating plasma parameters we have simultaneously measured the floating potential with tip 3 (Fig. 6) in a first run and the ion saturation current (Fig. 7) in a second one.

Figures 6 and 7 seem to indicate that the new probe reveals more details of the fluctuations due to turbulence than the simpler probe methods do. This result is likely because in these last cases the oscilloscope and the connecting cable load the probe pin cutting the high frequency components of the signals.

For more graphical clarity, the error bars associated with the measurements are not shown in the mentioned figures, but typical parameter errors from the $I-V$ characteristic fits are about 3% for the floating potential, 6% for the ion satu-

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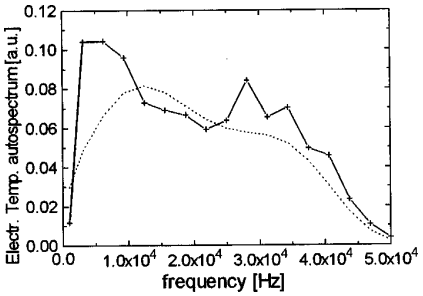


FIG. 9. The electron temperature autospectrum at 6 cm from the wall measured by the fast-sweep probe (−) with only eight samples and by the triple probe method (...) with 256 samples. For the fast-sweep probe, the $I-V$ characteristics are reconstructed with 150 points.

ration current, and 6% for the electron temperature (these last ones are instead shown in Figs. 8 and 11).

B. Electron temperature measurements

Figure 8 shows an example of the electron temperature measurements obtained by the fast-sweep probe method. The error bars of the displayed electron temperature indicate the uncertainties as given by the fitting procedure program.

A more appropriate experimental test of the goodness of the measure can be done by comparing the results with those obtained by the triple probe method. We realized triple probe configuration measurements with our probe by sampling the tip 3 floating potential V_3 and the tip 1 voltage V_1 (the so called return voltage) which is polarized at a positive potential with respect to pin 2 by a $V_b=9$ V floating battery. The tips 1 and 2 are connected as a floating symmetric double probe so that they draw the same current but with an opposite sign $I_1=-I_2$. This condition together with Eq. (1) applied to both tips separately can relate V_1 and V_b to V_f and T_e, but since $V_3=V_f$ the following relation is obtained:[2]

$$\frac{1-e^{-e(V_1-V_3)/kT_e}}{1-e^{-e(V_1-V_2)/kT_e}}=\frac{1}{2}, \tag{18}$$

if $V_2-V_1=V_b \gg kT_e/e$, which is valid in our experiment, then Eq. (18) simplifies to a direct voltages difference:

$$\frac{kT_e}{e}\approx\frac{V_1-V_3}{\ln 2}\approx\frac{V_1-V_3}{0.7}. \tag{19}$$

The above derivation is based on the assumption of space uniformity of the plasma parameters and a negligible reciprocal influence between the probe tips.

The time evolution of the fast-sweep probe electron temperature measurements cannot be compared directly with the corresponding one that results from a triple probe method, as this last method needs to use all the three tips of our probe and no tip remains for the fast-sweep probe. Anyway we can compare the power spectrum $S_{Te}(f)$ of the electron temperature fluctuations applying the digital correlation technique[14] to the time evolution data kept in different time. Figure 9 shows the estimation of $S_{Te}(f)$ as obtained from the two different methods for the same probe position and plasma discharge conditions.

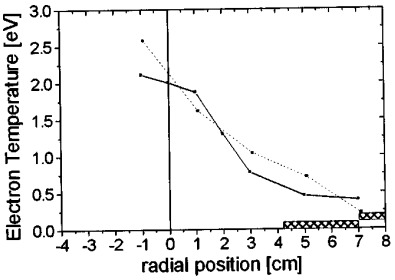


FIG. 10. The mean electron temperature as a function of the distance from the minor axis measured by the fast-sweep probe (−) and by the triple probe method (...).

The two measurements seem to give substantially two common features of the power spectrum: two broad band frequency peaks and a rapid fall at higher frequencies. The power spectrum measured from the triple probe is a smoother function of the frequency with respect to the other one because the number of the statistical ensemble realizations is quite large in the first case. The maximum data sample length of $2^{15}=32768$ byte limit our statistics for the power spectrum measured by the fast-sweep probe.

A meaningful comparison is also obtained by looking at the mean value and the standard deviation of the electron temperature fluctuations for different distances r from the minor axis of the toroidal chamber as is shown respectively in Figs. 10 and 11.

The fluctuation parameter profiles measured by the two methods are very similar for values of r comprised in the interval between 1 and 5 cm. The origin of the displayed discrepancies outside this region are attributed in the very edge of the plasma to the typical triple probe errors, and in the inner plasma to the fast-sweep prototype apparatus that has a sweep voltage range too small to scan the ion saturation region completely.

C. Electron temperature fluctuations and error bars

If we assume that the electron temperature fluctuations are quasinormally distributed around the mean value $\underline{T}_e$ with a centered second moment σ^2_{Te} (this is equivalent to assuming a weak correlation between the large number of the collective modes of the underlying plasma turbulence state) and

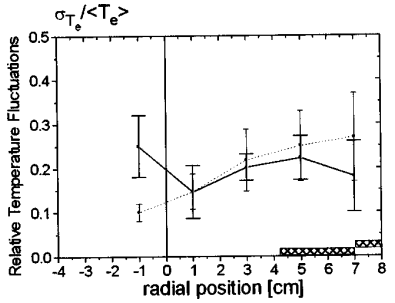


FIG. 11. Normalized electron temperature fluctuations as a function of the distance from the minor axis measured by the fast-sweep probe (−) and by the triple probe method (...). The error bars are estimated as explained in Sec. III C.

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the single measure $T_{e,j}$ is subject to experimental random errors with uncertainties $\Delta T_{e,j}$ then we expect $T_{e,j}$ to follow a Gaussian distribution again centered at $\underline{T}_e$ but with a square width $\underline{\sigma}_j^2 = \underline{\sigma}_{Te}^2 + \Delta T_{e,j}^2$ coming from the convolution of two Gaussian functions. We use as statistical estimators of the mean $\langle T_e \rangle$ and the square width σ_{Te}^2 of the electron temperature fluctuations the weighted mean with weight $w_j = 1/\Delta T_{e,j}^2$ of the N sampled data $T_{e,j}$:

$$\langle T_e \rangle = \frac{\Sigma_{j=1,N} w_j T_{e,j}}{\Sigma_{j=1,N} w_j}, \tag{20}$$

and the mean square with respect to $\langle T_e \rangle$ minus the mean square experimental error:

$$\sigma_{T_e}^2 = \frac{1}{N} \sum_{j=1,N} (T_{e,j} - \langle T_e \rangle)^2 - \frac{1}{N} \sum_{j=1,N} \Delta T_{e,j}^2. \tag{21}$$

By applying the error propagation law directly to Eqs. (20) and (21) we obtain the statistical errors of the estimated parameters which can be expressed by the following formula:

$$\left\{ \begin{array}{c} \Delta \langle T_e \rangle = \sqrt{\sum_{j=1,N} \left(\frac{\partial \langle T_e \rangle}{\partial T_{e,j}} \right)^2 \Delta T_{e,j}^2} = \frac{1}{\sqrt{\Sigma_{j=1,N} w_j}} \\ \Delta \sigma_{T_e} = \sqrt{\sum_{j=1,N} \left(\frac{\partial \langle \sigma_{T_e} \rangle}{\partial T_{e,j}} \right)^2 \Delta T_{e,j}^2 + \left(\frac{\partial \langle \sigma_{T_e} \rangle}{\partial T_{e,j}} \right)^2 \Delta \langle T_e \rangle^2} = \sqrt{\frac{\Sigma_{j=1,N} (T_{e,j} - \langle T_e \rangle)^2 \Delta T_{e,j}^2}{\sigma_{T_e}^2 N^2} + \frac{(\Sigma_{j=1,N} T_{e,j} - \langle T_e \rangle)^2}{\sigma_{T_e}^2 N^2 \Sigma_{j=1,N} w_j}} \end{array} \right\}. \tag{22}$$

It is important to notice that if we have a large data sample the error in the mean value can be kept arbitrarily small (if no source of systematic errors is present), the same is instead not true for the error of the second moment of the fluctuation distribution. As a practical rule the measure of $\underline{\sigma}_{Te}$ is statistically resolved if it is at least as large as the experimental error in the single measurement. The comparison of the two methods can then be performed only if we apply similar methods to calculate the error on the single sampled electron temperature measure.

The triple probe theory mentioned above considers the phase delay between two probe tips negligible, but the separation distance d cannot be reduced too much otherwise the magnetic shadow of one pin cover the other pin and hence introduce another error source. At the first order in d the relative difference of the generic plasma parameter as detected from the two tips is about equal to the phase difference $kd \cos\theta$ due to a passing wave having a wave number k and forming an angle θ with the straight line connecting the tips. Then the relative error in the electron temperature can be evaluated to be about $|k_y|_{\rm rms} d_{1,2}$. with $|k_y|_{\rm rms}$ being the root-mean-square poloidal wave number. In fact the observed electrostatic turbulence in the plasma immersed in a relatively strong magnetic field have very long parallel wavelength fluctuations and the dominant contribution to the error phase delay come from the poloidal separation between tip 1 and 2. The error bars shown in Fig. 11 for the triple probe plot correspond to the above described error estimation, in particular the value of $|k_y|_{\rm rms}$ is experimentally[15] about 2 $\rm cm^{-1}$ at the edge of the plasma and about 0.2 $\rm cm^{-1}$ in the inner plasma. Hence the error in the edge plasma is quite large. It is possible to correct the phase delay error using a fourth pin which permits us to reveal the difference V_1-V_3 simultaneously with the local poloidal wave number k_y of the fluctuations and thus compensate for the phase delay,[3] but we do not have a fourth tip available to do this correction.

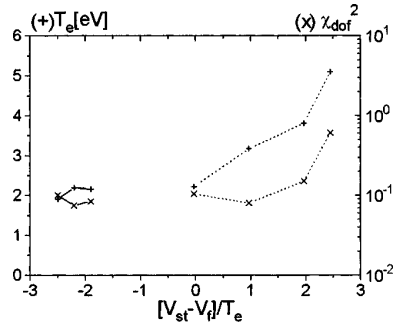


FIG. 12. The change of the electron temperature $T_e(+)$ and chi-square for degree of freedom $\chi^2_{\rm DOF}(x)$ with respect to the change of the starting voltage $V_{in}(-)$ and stopping voltage $V_{st}(\ldots)$ fit window. On the x axis the voltage V (which correspond to V_{in} in the curves on the left side and V_{st} in the curves on the right side) is referred to the floating potential V_f (obtained when the probe current is about 0 Å) and normalized with the electron temperature T_e obtained from the fit when $V_{in} = V_{\rm minimum}$ and $V_{st} = V_f$.

The contributions to the electron temperature error measurements for the fast-sweep apparatus could come from: (1) the frozen hypothesis of plasma parameters being invalid, (2) nonvalidity of the simple Langmuir probe model, (3) inaccuracy in the voltage and current measurements with the fast-sweep circuit. We make the strong hypothesis that all these sources of errors are included in the parameter fit error as obtained from the output of the fitting procedure based on the least square method. It is experimentally observed that from a single $I-V$ characteristic the parameter fit depends on the chosen time window (Fig. 4). In particular when the stopping voltage fit window V_{st} increases above the floating potential V_f the resulting electron temperature increases (Fig. 12) monotonically in agreement with the Langmuir probe behavior expected when a strong magnetic field is present.[16] Moreover the chi-square increases too (Fig. 12) and this can

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be considered an indication of a strong deviation from the assumed theoretical model expressed by Eq. (1). The fitting programs that have been used automatically change the starting and stopping voltages fit window and selects the trial fit having the minimum chi-square. This criterion gives fits that deviate randomly from fits with the imposed condition $V_{st} = V_f$ of about 7% in T_e and n but less than 4% in V_f.

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